Subhajit Dandapat

Curriculum Vitae

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About me

Computational Astrophysicist with extensive experience in large-scale data analysis, Bayesian inference, and source modeling for Gravitational-Wave (GW) research. Currently advancing **AI**-driven and **Machine-Learning** methodologies for GW data analysis, simulation-based inference, and physics-informed modeling. Active collaborator across major international consortia, including the International Pulsar Timing Array (IPTA) consortia, Indian Pulsar Timing Array (InPTA), Laser Interferometer Space Antenna and have previously worked with the LIGO-Virgo-KAGRA Collaboration (LVK). Passionate about leveraging AI and high-performance GPU computing to advance data-driven scientific discovery and develop scalable, interpretable learning models for complex physical systems.

Education

2024–current **Postdoctoral Research Fellow**, *Department of Physics, National University of Singapore, Singapore*

Advisor: Dr. Alvin Chua

2019–2024 **Doctor of Philosophy in Astrophysics**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*

Thesis title: *Modeling Nano and Hecto Hertz Gravitational Wave sources and their observational implications*

Advisor: Prof. A. Gopakumar

2014–2019 **BS-MS Dual Degree in Physics, Indian Institute of Science Education and Research**, Bhopal, Madhya Pradesh, India

MS Thesis title: *Chiral Anomalies in Quantum Field Theory*

Studied: *Physics (Major), Mathematics, and Chemistry*

CGPA:8.89/10

Professional Experience

2024-Present
- **External Attention Transformer with Physics-Informed Positional Encoding** – Built a novel PyTorch framework for detecting Eccentric supermassive black hole binaries (SMBHBs) in highly heterogeneous multimodal pulsar timing array (PTA) datasets. Outperforms traditional positional encodings by accurately extracting signal parameters from complex noisy time-series, blending astrophysical modeling with modern ML architectures.
- **Amortized Bayesian Inference with Continuous Normalizing Flows (CNFs) & Diffusion Models** – Trained neural surrogates to approximate Bayesian posteriors for high-dimensional GW data. Implemented CNFs (invertible mappings) and diffusion models (denoising dynamics) for amortized inference, enabling scalable posterior sampling, accelerating uncertainty quantification, and reducing reliance on costly MCMC; exploring extensions with physics-informed transformer encodings to further enhance expressivity and data efficiency.
- **Optimizing GW Signal Search with Physics-Informed Neural Networks (PINNs)** – Applied PINN surrogates to replace costly ODE/PDE solvers in International PTA pipelines for inspiraling, eccentric, spinning SMBHBs; accelerated Bayesian likelihood evaluations, reduced computational bottlenecks, and enabled scalable, hardware-accelerated GW data analysis.

2022-2024
- **Bayesian Inference with Large-Scale MCMC** – Applied high-dimensional (several hundreds of parameters) MCMC Bayesian inference to extract weak GW signal parameters buried in heterogeneous, noise-dominated pulsar timing array (PTA) datasets; enabled detection and characterization of supermassive black hole binaries (SMBHBs) from real observational data.

Selected Publications

[1] Dandapat, S. et al. (2024). *Efficient prescription to search for linear gravitational wave memory from hyperbolic black hole encounters and its application to the NANOGrav 12.5-year dataset*. Phys. Rev. D 109, 103018. arXiv:2402.03472.

[2] Agazie, G., Dandapat, S., et al. (2024). *Comparing recent PTA results on the nanohertz stochastic GW background*. Astrophys. J. 966, 105. arXiv:2309.00693.

[3] Dandapat, S. et al. (2023). *Gravitational waves from black-hole encounters: Prospects for ground and galaxy-based observatories*. Phys. Rev. D 108, 024013. arXiv:2305.19318.

[4] Cho, G., Dandapat, S., & Gopakumar, A. (2022). *Third order post-Newtonian gravitational radiation from two-body scattering: Instantaneous energy and angular momentum radiation*. Phys. Rev. D 105, 084018. arXiv:2111.00818.

For a detailed list of publications, please refer to my Google Scholar page.

Leadership Roles

2023 - present
- Supervising NUS graduate (Mark John) and PhD students (Vidit Singh, TIFR Mumbai; Adya Shukla, IIT Roorkee) on projects involving gravitational-wave data analysis and AI-based approaches.
- Co-led the **IPTA Data Release 3 Search** for Gravitational Wave Burst with memory project.
- Led quantitative comparisons of GW posteriors across multiple **international PTA** collaborations by evaluating distribution of their differences, moving beyond traditional point-based metrics (e.g., Mahalanobis distance) to more rigorous distribution-level comparisons. (Arxiv:2309.00693; Arxiv:2306.16225 & Arxiv:2306.16214).

2022 - 2023
- Served as **Deputy Managing Leader (DML)** in Indian PTA, overseeing data reduction and backups for uGMRT datasets.

Skills

Programming Languages	`Python, Wolfram Language, Bash, C, LaTeX`
Machine Learning & AI	`PyTorch, Scikit-learn, NumPy, SciPy, Pandas, Normalizing Flows, PINNs, Transformer Simulation-based Inference`
Computing Software	`MATHEMATICA, MAPLE`
Astrophysical Software	`ENTERPRISE, BILBY, TEMPO2, PINT`
Telescope Observations	Upgraded Giant Metrewave Radio Telescope (uGMRT)
Data Analysis	Machine Learning, Bayesian Inference, Time-series Analysis, Data Visualization
High-Performance Computing	Parallel Processing, GPU Computing (A40, H100), PBS/Slurm Environments

AI Grants Writing & Computing Resource Awards

- **Principal Author**, *CFP03 HPC Project Resource Allocation Award (NUS HPC, 9 Oct 2025 – Oct 2026)* — **Approved** computing allocation of **30K NVIDIA H100 GPU-hours**, **2K NVIDIA A40 GPU-hours**, and **200K CPU-hours** for the project *"AI- and GPU-Accelerated Search for Gravitational Waves in PTAs."*
- Co-authored **NRF Singapore AI-for-Science (AI4S) Catalytic Grant** – *"Unified AI Solutions to Enable Scientific Inference for Next-Generation Gravitational-Wave Astronomy"*. 05 September 2025, **1.5M SGD**, Singapore National Research Foundation.
- Contributed to **NAII Seed Grant** – *"Using Artificial Intelligence to Enhance Bayesian Inference in Gravitational-Wave Astronomy"*. 15 April 2025, **250k SGD**, NUS Artificial Intelligence Institute FY2025 H1 Seed Grant Call.

Fellowships

2025 NANOGrav **AccelNet** Travel Grant for international travel to attend conferences and research visits in the USA, June 15 - July 4, 2025.

2023 **Sarojini Damodaran Fellowship** from TIFR supporting international travel to Europe for conferences and research visits.

2022 **Pauli Center for Theoretical Studies visitor program Fellowship**, ETH and University of Zürich, July 15 - September 17, 2022.

Seminars and Conference Presentations

2025
- **"Searching for Linear GW Burst with memory in IPTA DR3 Dataset"**; In person talk on IPTA 2025 Science Meeting, **Caltech, Pasadena, US**; June 27, 2025.

2023
- **"Detection and Characterization of Burst with Linear Memory Events with Pulsar Timing Array and LIGO-Virgo-KAGRA Observatories"**; In person talk on Astrophysics group of Milan-Bicocca; December 5, 2023.
- **"Characterizing Burst with Linear Memory Events with LIGO-Virgo-KAGRA and Pulsar Timing Array Observatories"**; Amaldi15 virtual conference; July 17-21, 2023. [YouTube link]
- **"Searching Burst with memory event in IPTA dataset and Compairing PTA posteriors"**; IPTA Science meeting; 19-23 June, 2023.
- **"Comparing PTA posteriors"** IPTA GWA hackweek on 3P+ comparisons virtually on zoom; 6-8 March, 2023.

2022 **"Stochastic gravitational wave background spectrum due to supermassive black hole binaries in precessing eccentric orbits"**; Gravitational Wave Orchestra in-person held on UCLouvain, Belgium; Sep 8-9,2022.

2021
- **"Modeling GW burst with linear memory events"**; EPTA 2021 Winter meeting virtually on zoom Dec 6-8, 2021 and Astronomy Society of India meeting in-person held on **IIT Roorkee**; Mar 25-29, 2021.
- **"Effect of relativistic pericentre advance on the SGWB due to eccentric SMBH binaries"**; EPTA 2021 Summer meeting virtually on zoom; Apr 21-23, 2021.

Research Visits

2025 Visited Prof. Maura McLaughlin's group at West Virginia University from **30 th June to 3rd July, 2025** for IPTA collaborative work.

2023
- Visited Prof. Alberto Sesana's B Massive group at Milano-Bicocca from **4 th December to 8th December, 2023** for IPTA collaborative work.
- Visited Prof. Jetzer's group at University of Zürich from **20 th November to 1st December, 2023** for LVK collaborative work.

2022 Visited Prof. Jetzer's group at University of Zürich from **15 th July to 30th September, 2022** for LVK collaborative work.

References

Available on Request